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MACHINE LEARNING (25SC2107E)

Y25 - 2026-2027 Trimester - 4

ABSTRACT

Sec No-1	Team No-16	Project Title- SMS and Email Spam Filter

1.Abstract

The SMS and Email Spam Filter is a machine learning-based system developed to automatically identify and filter unwanted, irrelevant, or suspicious messages. With the increasing use of SMS and email for communication, users are exposed to a large number of spam messages, making it difficult to identify them manually. The main objective of this project is to classify messages into two categories: Spam and Not Spam, based on the content and patterns present in the message.

The system uses a labelled dataset containing examples of both spam and legitimate messages. The collected text data is first processed to make it suitable for analysis. Bag-of-Words and TF-IDF (Term Frequency-Inverse Document Frequency) techniques are used to convert the text into numerical features that can be understood by machine learning algorithms. These features help the system identify important words and patterns that are commonly associated with spam messages.

Machine learning algorithms such as Multinomial Naive Bayes and Support Vector Machine (SVM) are used to train the classification model. After training, the model can analyse new and unseen SMS or email messages and predict whether they are spam or legitimate. The project also focuses on comparing different text representation and classification approaches to understand which method performs better. In addition, the influential words or terms that contribute to the classification can be studied to better understand how the model makes its predictions.

The proposed system provides a practical approach for reducing unwanted messages and improving the efficiency of message filtering. It also demonstrates the application of Machine Learning, Natural Language Processing, text preprocessing, feature extraction, and text classification to a common real-world problem. Overall, the project aims to develop a simple and effective spam filtering system while providing a clear understanding of how machine learning can be applied to textual data.

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